2024-25 CSC357 Brain-Inspired Artificial Intelligence

Coursework PART A

February 2025

NOTE:

- You need to submit a brief technical report based on the three lab tasks from Weeks 1-5.
- This document constitutes PART A of the CSC357 Coursework. Both Coursework PART A and PART B should be submitted together in a single file (PDF or Microsoft Word).
- PART A accounts for 10% of the total assessment for the module.

1. Task 1 - Neural modelling (Mark 3%)

This task requires you to familiarize yourself with the Integrate and Fire (IF) model, its Python implementation, and the effects of model parameters.

- Download `CSC357_Lab2_neural_modelling.ipynb` from Canvas (located in Week 2 module content).
- Open the downloaded file using Jupyter Notebook (use Google Colab if having problems with local Python installation).
- Go through the lab content. Understand how the IF model is implemented using the Euler method.

Answer the following two questions. Include your answers in your coursework submission.

Q1 Maintain all other parameters at their default values and run the model simulation with lower values of I_{inj}. Can you identify and report the critical I_{inj} value that fails to produce any spike? In your coursework, **include a copy of the membrane potential trace figure** generated from `plot_volt_trace()`, simulated under the critical I_{inj} value. [Mark 2%]

Q2 Maintain all other parameters at default values (including $I_{inj} = 230$), and run the model simulation with different refractory periods. You can do so by re-initializing the parameter, e.g., `pars = default_pars(T=1400, tref=5)`. In your coursework, **provide a statement on how the increase in the refractory period influences the number of spikes generated per unit of time.** [Mark 1%]

2. Task 2 - Features in neural networks (Mark 4%)

This lab will examine visual features generated from a convolutional neural network and how they are used for object recognition (or classification).

- Download `CSC357_Lab3_neural_networks.ipynb` from Canvas (located in Week 3 module content).
- Open the downloaded file using Jupyter Notebook (use Google Colab if having problems with local Python installation).
- Go through the lab content. Understand how convolutional filters are applied to the input image and how activation maps are calculated at different layers.

Answer the following three questions. Include your answers in your coursework submission.

Q1 Find a different test image. Load (or download) and preprocess your own test image using the provided code. **Include a copy of your test image and its the classification result. Additionally, comment on the success of the pre-trained AlexNet on your test image** (i.e., did the model yield the correct result?) [Mark 2%]

Q2 In your coursework, **include the activation maps from all convolutional layers. Provide a discussion on the significant regions at different layers that the model relies on to make label predictions.** [Mark 1%]

Q3 Select a close-up portrait image. In your coursework, **include the portrait image along with its classification results from the model. Evaluate whether the results align with your own categorization. If not, discuss potential reasons for the discrepancy.** [Mark 1%]

3. Task 3 - Decision and decision time (Mark 3%)

This lab will fit a computational model (i.e., the drift-diffusion model) to perceptual decision-making behaviour based on your experimental results.

- Download `CSC357_Lab4_decision_making.ipynb` from Canvas (located in Week 4 module content).
- Open the downloaded file using Jupyter Notebook (Recommend to use Google Colab since you need to install a few new libraries).
- If you are conducting this lab on Google Colab, you can leverage GPU acceleration to expedite the fitting procedure. To enable this, navigate to `Runtime` - `Change runtime type` in the menu, and select `GPU` under `Hardware accelerator`. Note that there is a GPU usage limit on Colab.
- Go through the lab content. In this lab, you first need to complete a behavioural experiment online. Afterwards, you download your dataset for subsequent analyses.

Answer the following two questions. Include your answers in your coursework submission.

Q1 Complete the perceptual decision-making task online. You can start the task via [this link]. Section 1 of the lab notebook provides further information about the experiment. After the experiment, download and preprocess your data, using codes in Section 2 as examples.

In your coursework, include the following results [Mark 1%]:

- A histogram showing the response time distribution.
- Mean accuracy and response time in the easy and difficult conditions, separately.
- A short statement on the behavioural difference (if any) between the easy and difficult conditions.

Q2 Define and fit a drift-diffusion model to your behavioural data (as in Section 3 of the notebook).

In your coursework, include the following results [Mark 2%]:

- Posterior model prediction plots together with the observed data (as in Section 3.2)
- The means and standard deviations of the fitted model parameters (as in Section 3.3)
- Posterior distributions of the drift rate estimates from the easy and difficult conditions
- A short statement on how task difficulty influences the drift rate.